Module 3 · Temporal Knowledge & Graph Management

POSTGRESQL TRANSACTION · SQLx · namespace lock + live job lease

Module 2 claims
Maya / home city / Bengaluru
correction:true · confidence:0.95

claims + quotes
+ optional vectors

Apply guard
Revalidate · check lease

Maya · person
namespace: user:maya

Entity resolver
Exact key → unique alias → new

entity ID + home city

Existing value?
subject + predicate

Same value

Reinforce
Reuse assertion ID

No prior value / non-single

Create
New active assertion

Different value · single slot
Chennai → Bengaluru

Residence validity
R1 Chennai: [1 Oct, 10 Oct) · superseded
R2 Bengaluru: [10 Oct, …) · active
recorded_at is retained separately.
R2 → C3 + E3 quote · R2 → Maya / extracted mentions.

Reviewed correction ≥ 0.8?

Yes

Supersede
R1 Chennai → superseded
R2 Bengaluru → active

R2 → supersedes → R1

Authoritative store
entities
assertions + vectors
assertion_sources
assertion_entities
assertion_edges

No

Contest
Old + new → contested

contradicts

Module 2 observation
"Maya works remotely as a software engineer at NovaTech."
supports: W1, W2, W3

apply_consolidation
lock + lease check

Supports still valid?
2+ active IDs
grounded subject

Yes

Observation writer
Reuse or insert
Copy evidence + entity links

No

Rollback / retry

Changed support / retraction

O1 · kind: observation
O1 → supports → W1 / W2 / W3

Dependency invalidator
Dependent observations → stale

queue affected IDs

memory_jobs
project: {assertion_id: R2}
consolidate: {episode_id: P2}

COMMIT · assertion changes + evidence links + jobs + job completion

Committed job records

consolidate P2

project R2 → snapshot
properties + entities + relations + revision

Projection worker
Load one MVCC snapshot
Revision-guarded upsert

Cypher / HTTP
reject older revisions

Neo4j projection
Assertion
KnowledgeEntity
MENTIONS · SUPPORTS

graph candidates

Projected relationships

Module 2
Consolidation

R2
Bengaluru · active

MENTIONS

Maya
KnowledgeEntity

SUPPORTS
relation: supersedes

R1
Chennai · superseded

Module 4
Recall + graph expansion

PostgreSQL remains authoritative.
Quotes stay in PostgreSQL.
Module 4 rechecks graph hits against stored status and time.

Maya data and IDs are illustrative.
Entity matching uses namespace + type + normalized name.